

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained a **customized Linux operating system** optimized for Kubernetes, including kernel enhancements for scalability and reliability, supporting a fleet of hundreds of clusters.
- Built high-performance diagnostic tools using **Go and Python** to automate infrastructure tasks and troubleshoot system-level issues across millions of compute instances.
- Implemented **BPF-based network segmentation** and service mesh solutions, integrating with Cilium for enhanced security and traffic management within the Kubernetes platform.
- Managed the full lifecycle of hundreds of Kubernetes clusters, including **provisioning, OS and Kubernetes upgrades**, and decommissioning, ensuring seamless transitions and minimal disruption.
- Designed and implemented **robust observability frameworks** using tools like Splunk and Dynatrace for system health monitoring, performance assurance, and proactive issue resolution at scale.
- Collaborated with cross-functional teams to validate, adopt, and integrate optimized Linux OS distributions across diverse infrastructure environments, ensuring compatibility and performance.
- Containerized infrastructure services using **Docker** and deployed them onto **AWS EC2** instances managed via Jenkins CI/CD pipelines, improving deployment efficiency and consistency.

SKILLS

Operating Systems – Linux, Linux Kernel, Shell Scripting

Containerization & Orchestration – Kubernetes, Docker, Cilium, BPF

Programming Languages – Go, Python, C++

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

CI/CD & Version Control – Jenkins, Git

Observability & Monitoring – Splunk, Dynatrace, CloudWatch

PROJECTS

Kubernetes Cluster Automation Platform

- Developed Python scripts to automate the provisioning and configuration of **Kubernetes clusters** on AWS, reducing setup time by 60%.
- Engineered a Go-based service for managing cluster upgrades, implementing rolling update strategies that maintained 99.95% availability for critical applications.
- Integrated **Docker** containerization for all cluster management components, ensuring consistent deployment across development and production environments.

Linux OS Hardening for Containerized Workloads

- Customized the Linux kernel configuration for enhanced security and performance, focusing on process scheduling and memory management for containerized applications.
- Implemented system-level security controls and auditing mechanisms to meet rigorous **Linux security** requirements for enterprise infrastructure.
- Authored comprehensive documentation and runbooks for the hardened OS, enabling smoother adoption by development teams and reducing onboarding time by 30%.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)